

Algebra First-Year Exam, January 2014, Part 1

Answer four problems. (If you turn in more, the first four will be graded.) Within reason, you may quote theorems as long as you state them clearly.

1. Give an example of each of the following.
 - (a) A non-trivial abelian group.
 - (b) A non-abelian simple group.
 - (c) A non-nilpotent solvable group.
 - (d) A non-abelian nilpotent group.
2. Let G be a group and let $g \in G$.
 - (a) Define what is meant by the order of g .
 - (b) Let g have infinite order, and suppose that N is a finite normal subgroup of G . Prove, from your definition, that the element $gN \in G/N$ has infinite order.
3. This problem has two parts.
 - (a) Define what it means for a subset of a group G to generate G .
 - (b) Let G be a group, and suppose $|G| = 2^n$. Prove that there is a subset S of cardinality n such that S generates G .
4. State and prove the Second Isomorphism Theorem for groups.
5. This problem has two parts.
 - (a) Let S be a simple group and suppose that S acts non-trivially on a finite set A with $|A| = n$. Prove that $|S| \leq n!$.
 - (b) Suppose $G = A_5$ (the alternating group on five letters) is acting on a set A in such a way that no element of A is fixed by every element of G . Suppose that $|A| = 10$. Prove that G has at most two orbits in its action on A .
6. State and prove the First Sylow Theorem.